Thabo Mantsima

Mechatronics Engineer | Backend & Embedded Software Developer

+267 72 470 427 • mantsimat@gmail.com • Gaborone, Botswana • linkedin.com/in/thabo-mantisma • github.com/BLKamau

PROFILE

Recently graduated Mechatronics and Industrial Instrumentation engineer (BIUST, 2026) with a hybrid engineering and software background and a lean toward backend development. Combines six months of hands-on industrial instrumentation and control experience at a 600MW power station with independent software projects spanning embedded systems, computer vision, and backend web development — including a shortlisted Banking-as-a-Service platform built for the FNB Hackathon. Comfortable moving between control-system troubleshooting, hardware integration, and building production-style backend services.

TECHNICAL SKILLS

Backend & Web: Python, Django, Django REST Framework, Django Ninja (REST APIs), Streamlit, HTMX, Tailwind CSS, Alpine.js, SQL/relational database design (PostgreSQL), Git, Railway deployment

Frontend (developing): React, Vite, JavaScript

Embedded & Hardware: Raspberry Pi 4/5, Hailo-8 AI HAT, Coral Edge TPU, KiCad schematic design, sensor integration (LiDAR, IMU), computer vision (YOLOv8n, MiDaS, DeepLabV3+)

Instrumentation & Control: DCS, SCADA, PLC programming logic, HART communicator, HMI, CEMS, field & lab instrument calibration (DPT, RTD, thermocouples, flow/level transmitters)

Engineering & CAD/CAM: FANUC G-code (CNC lathe/mill), APT part programming, robotics (DH convention, Jacobians), systems architecture, technical documentation

PROFESSIONAL EXPERIENCE

Mechatronics Engineering Intern — Instrumentation Maintenance Department Feb 2025 – Jul 2025

Morupule B Power Station, Botswana Power Corporation

600MW coal-fired circulating fluidized bed power plant

- Performed preventive and corrective maintenance, and field/lab calibration of pressure, level, and temperature instrumentation (DPTs, RTDs, thermocouples, flow/level transmitters) using HART communicators and precision gauge comparators.
- Diagnosed and corrected a malfunctioning differential-pressure level sensor on a demineralized water tank, improving measurement accuracy from ±8.8% to ±1.0% of span (an 87% improvement).
- Root-caused and realigned a misaligned infrared dust analyzer on the Continuous Emissions Monitoring System (CEMS), correcting the displayed CEMS value from 45% to 90% of specification.
- Tested and repaired electric actuator control systems — three-phase motor phase-balance testing, control/display/power card fault diagnosis, and position-limit configuration — under senior technician supervision.
- Led a control panel rewiring project for the Air Duct Burner System, documenting and reorganizing tangled wiring into a traceable cable management system that cut troubleshooting time from hours to under 30 minutes.
- Rotated through Water Treatment, Coal/Limestone/Ash Handling, Air Handling, Electrical Maintenance (HVAC, UPS, high-voltage protection), and Condition-Based Monitoring departments, gaining exposure to DCS, SCADA, PLCs, and industrial communication protocols (HART, Modbus, Ethernet).
- Supervised wagon tippler operations and coordinated instrument calibration assignments for fellow technicians during peak workload periods.

Backend Developer (Freelance) Present

Enovative Inc. — Creative Startup Agency

- Co-developed and deployed the agency's own profile/landing page.
- Contributes across software development, system architecture, and client-facing technical work, including a two-path pricing quotation and Software Requirements Specification (SRS) package delivered for a client's custom job application platform, with swim-lane process diagrams and BWP-denominated pricing.

Backend & Systems Integration — Phantom Banking 2025

FNB Hackathon

- Co-developed Phantom Banking, a Banking-as-a-Service (BaaS) platform enabling businesses to serve unbanked customers by spawning sub-accounts/wallets under their FNB merchant profile, with optional KYC upgrades to full accounts.
- Owned wallet logic and backend orchestration on a Django + Django REST Framework + PostgreSQL backend, integrating with a React (Vite) + Tailwind CSS frontend to support QR and EFT payment acceptance across embedded wallets.
- Contributed to API documentation and testing tooling (Swagger, Postman) and process mapping (BPMN, Camunda-ready) as part of a 5-person cross-functional team.

KEY PROJECTS

- NavSense — Wearable navigation system for visually impaired users built on Raspberry Pi 5 with a Hailo-8 AI HAT, YDLiDAR SDM18, and BNO055 IMU; features semantic node mapping, a tiered vision pipeline with graceful degradation, bilingual (English/Setswana) guidance, and bone-conduction haptic feedback. Reached a working end-to-end prototype demo, building on an earlier iteration (Raspberry Pi 4B, Coral Edge TPU, YOLOv8n/MiDaS/DeepLabV3+ five-layer vision architecture). Includes fully built prototype.
- Reinforcement Learning Pong Agent — Modular Python RL project training agents to play Pong, built on a custom Pygame game wrapped as a Gymnasium-compatible environment; supports PPO/DQN training, multiple observation types (low-dimensional and pixel-based), and a full pipeline for training, evaluation, and analysis (benchmarking, agent comparison, training-curve visualization), plus an interactive menu system with timed game modes.
- Remote Patient Monitoring Telemetry Pipeline (BIUST TELE 523) — System architect for a Python/Streamlit pipeline simulating modulation schemes, channel effects, and feature extraction on the Parkinson's Telemonitoring Dataset, with a three-tier alert system and six-page live dashboard.
- Personal Engineering Portfolio — Django site (Django Ninja API, Tailwind CSS, HTMX) showcasing engineering and software projects, deployed on Railway.

EDUCATION

BEng Mechatronics and Industrial Instrumentation

Botswana International University of Science and Technology (BIUST) — 5-year program, graduated July 2026

- Relevant coursework: CNC/CAM (FANUC G-code, APT programming, Bezier curve tangency), robotics (DH convention, Jacobians), materials science (Weibull reliability, DFMA), data communications, drone systems design.